

Product design specifications (PDS):

◆ 1. Function

The system shall:

- *Lift the ball to a specified height*
- *Identify the ball color*
- *Sort/direct the ball based on color*
- *Shoot the ball toward a designated target*
- *Calculate and display the score*

◆ 2. Performance Requirements

- *Small Ball detection response time*
- *Lifting time*
- *Color detection high accuracy*
- *System cycle time*
- *Product Life Span: Expected lifespan of the system is 3-5 years with proper maintenance. The servo motors and stepper motors are expected to last for at least 1,000 cycles, while the sensors may require calibration and cleaning after every 500 cycles.*

◆ 3. Physical Characteristics

- *Maximum dimensions: 100 x 100 cm*
- *Ball size: 20 mm*

◆ 4. Inputs and Outputs

Inputs:

- *Ball presence (IR/proximity sensor)*
- *Ball color (color sensor)*

Outputs:

- *Motor control signals*
- *Shooter activation*
- *Score display (LCD / 7-segment)*

◆ 5. Control System

- *Arduino*
- *Real-time processing of sensors*
- *Decision-making logic for sorting and shooting*
- *Score calculation algorithm*

◆ 6. Electrical Requirements

- *Supply voltage*
- *Power consumption*
- *Safe wiring and insulation required*

◆ 7. Mechanical Requirements

- *Stable lifting mechanism (conveyor)*
- *Reliable sorting mechanism (servo gate or different paths)*
- *Consistent shooting force (spring or motor-driven)*

◆ 8. Reliability

- *Replaceable components (motors, sensors)*

◆ 9. Cost Constraints

- *Total budget*
- *Use of available and low-cost components preferred*

◆ 10. Testing

- *Sensor calibration tests*
- *Accuracy testing (color & shooting)*
- *System integration test*

◆ 11. Constraints

- *Use standard electronic components*
- *Comply with lab/project guidelines*